Twin Layer: Sensor Specification

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Purpose and scope

This document fixes every sensor ID, the JSON each sensor returns and the rules the Twin applies to its values, so the Twin, City Brain, Guardian and dashboard teams build against one contract.

It covers 16 sensors across four zones (Port, Industrial, Downtown, Residential) and the 8 actuators and traffic lights they connect to. Every ID, range and interval is a proposed starting point for the simulation, not a real-world calibration, and must be confirmed against the challenge specification before implementation.

ID convention

Every device ID follows `<ZONE>-<TYPE>-<NN>`; for example, `PRT-WLV-01` is the first water level sensor in the Port.

Zone code	Zone
PRT	Port
IND	Industrial
DTN	Downtown
RES	Residential

Type codes and the unit each device reports:

Type code	Device	Output unit
WLV	Water level sensor	cm
FLW	Flow sensor	L/s
PRS	Pressure sensor	bar
TMP	Temperature sensor	°C

Type code	Device	Output unit
TRF	Traffic counter	veh/min
PMP	Pump (actuator)	on/off, speed %
VLV	Valve (actuator)	open %
TLT	Traffic light (actuator)	phase

Rules:

- IDs are uppercase, fixed once created and never reused, even after a device is removed.
- `NN` is a two-digit counter per zone and type, starting at 01.
- A replacement device gets a new ID and a `replaces` field pointing to the old ID, so history stays linked.

Sensor registry

The Twin registers 16 sensors, four per zone; the critical readings (Port water level, Industrial temperature) have a backup sensor.

Sensor ID	Measures	Unit	Normal range	Hard limits	Interval	Linked device	Backup sensor
PRT-WLV-01	Harbour water level	cm	20–80	0–300	5 s	PRT-PMP-01	PRT-WLV-02
PRT-WLV-02	Harbour water level (backup)	cm	20–80	0–300	5 s	PRT-PMP-02	PRT-WLV-01
PRT-FLW-01	Pump outlet flow	L/s	0–250	0–500	5 s	PRT-PMP-01	none
PRT-PRS-01	Pump header pressure	bar	2–6	0–10	5 s	PRT-PMP-01	none
IND-TMP-01	Process temperature	°C	20–90	-10–150	10 s	IND-VLV-01	IND-TMP-02
IND-TMP-02	Process temperature (backup)	°C	20–90	-10–150	10 s	IND-VLV-01	IND-TMP-01
IND-PRS-01	Coolant line pressure	bar	2–6	0–10	10 s	IND-VLV-01	none

Sensor ID	Measures	Unit	Normal range	Hard limits	Interval	Linked device	Backup sensor
IND-FLW-01	Coolant line flow	L/s	10–120	0–200	10 s	IND-VLV-01	none
DTN-TRF-01	Vehicles, intersection A	veh/min	0–60	0–120	10 s	DTN-TLT-01	none
DTN-TRF-02	Vehicles, intersection B	veh/min	0–60	0–120	10 s	DTN-TLT-02	none
DTN-WLV-01	Storm drain level	cm	0–60	0–200	5 s	DTN-VLV-01	none
DTN-TMP-01	Ambient temperature	°C	5–35	-10–50	60 s	none	none
RES-WLV-01	Reservoir level	cm	100–250	0–300	10 s	RES-VLV-01	none
RES-PRS-01	Network pressure	bar	3–5	0–10	10 s	RES-VLV-01	none
RES-FLW-01	Network flow	L/s	5–60	0–150	10 s	RES-VLV-01	none
RES-TRF-01	Vehicles, main road	veh/min	0–40	0–100	10 s	RES-TLT-01	none

Normal range colours the dashboard; hard limits clamp every value the Twin produces.

Output format

Every sensor reading is one JSON object with the same fields, whether it comes from an MCP tool, the live stream or the event history.

```
{
  "sensor_id": "PRT-WLV-01",
  "zone_id": "PRT",
  "type": "water_level",
  "unit": "cm",
  "timestamp": "2026-10-18T17:30:05Z",
  "sim_tick": 1042,
  "seq": 88213,
  "actual_value": 40.0,
  "reported_value": 180.0,
  "delta": 140.0,
  "status": "OK",
  "fault": "FALSE_READING",
  "available": true,
  "backup_ids": ["PRT-WLV-02"],
  "linked_device_ids": ["PRT-PMP-01"]
}
```


Field	Type	Meaning
sensor_id	string	Unique ID from the registry
zone_id	string	PRT, IND, DTN or RES
type	string	water_level, flow, pressure, temperature or traffic_count
unit	string	Fixed by type: cm, L/s, bar, °C, veh/min
timestamp	string	ISO 8601, UTC, simulated clock
sim_tick	integer	Simulation step that produced the reading
seq	integer	City-wide counter that only increases; used for replay and ordering
actual_value	number	True simulated value
reported_value	number or null	What the sensor reports; null when it sends nothing
delta	number or null	reported_value minus actual_value; no verdict attached
status	string	OK, DEGRADED, OFFLINE or FAILED

Field	Type	Meaning
fault	string or null	Active fault type, or null
available	boolean	False when the sensor cannot be used
backup_ids	array	Compatible backup sensors, may be empty
linked_device_ids	array	Actuators this sensor feeds

When a sensor is down, the Twin sends null, never 0:

```
{
  "sensor_id": "IND-TMP-01",
  "zone_id": "IND",
  "type": "temperature",
  "unit": "°C",
  "timestamp": "2026-10-18T17:30:10Z",
  "sim_tick": 1043,
  "seq": 88240,
  "actual_value": 74.5,
  "reported_value": null,
  "delta": null,
  "status": "FAILED",
  "fault": "DEAD",
  "available": false,
  "backup_ids": ["IND-TMP-02"],
  "linked_device_ids": ["IND-VLV-01"]
}
```

Numbers use up to two decimals. Which callers may see `actual_value`, `delta` and `fault` is still to be decided (see Open decisions).

Dual-state values and faults

`actual_value` always follows the simulation, and `reported_value` equals it unless a fault is active; for example, actual water level 40 cm, reported 180 cm.

Fault type	Effect on reported_value	Resulting status	Parameters
FALSE_READING	Fixed value, or offset from actual	OK	value or offset
STUCK	Frozen at the last value before the fault	OK	none
DRIFT	Offset from actual grows every tick	OK	rate per tick
NOISE	Random jitter around actual	OK	amplitude
DELAYED	Shows the actual value from N ticks earlier	DEGRADED	lag_ticks
DROPOUT	null, no new readings, may come back	OFFLINE	duration_ticks (optional)
DEAD	null until repaired or replaced	FAILED	none

A false, stuck or drifting sensor still shows status OK, because from the sensor's side nothing is wrong. The Twin only exposes the discrepancy; Guardian decides whether it is a security issue.

Faults are started with `simulate_device_failure` or `run_scenario`, checked with `get_device_status`, and cleared with `reset_scenario` or by registering a replacement device.

Status values and error codes

A sensor has one of four statuses, and every failed request returns one of the error codes below.

Status	Meaning
OK	Readings arrive on time
DEGRADED	Readings arrive but late or noisy
OFFLINE	No readings now; may return

Status	Meaning
FAILED	Device is dead; needs repair or replacement

Error code	Meaning	Origin
DEVICE_UNAVAILABLE	Device is failed or offline	Architecture proposal
NO_COMPATIBLE_DEVICE_AVAILABLE	No backup device exists or is free	Architecture proposal
SENSOR_NOT_FOUND	Unknown sensor ID	New
INVALID_FAULT_TYPE	Fault name not in the fault table	New
FAULT_ALREADY_ACTIVE	Sensor already has a fault	New
NOT_AUTHORIZED	Caller may not do this	New
GUARDIAN_RESTRICTED	Guardian has restricted this device	New

The Twin raises no alarms and chooses no response: normal ranges only colour the dashboard, and priorities stay with the City Brain.

MCP responses and events

Every MCP tool wraps its result in the same envelope, so callers check `ok` first.

`get_device_status(sensor_id)` returns one reading object:

```
{
  "ok": true,
  "data": { "sensor_id": "PRT-WLV-01", "reported_value": 180.0, "status":
"OK" },
  "timestamp": "2026-10-18T17:30:05Z"
}
```

`get_city_state(zone_id?)` returns the sensors and actuators of one zone, or of all four when `zone_id` is omitted:

```
{
  "ok": true,
  "data": {
    "zone_id": "PRT",
    "sim_tick": 1042,
    "sensors": [ { "sensor_id": "PRT-WLV-01", "reported_value": 180.0,
"status": "OK" } ],
    "actuators": [ { "device_id": "PRT-PMP-01", "status": "FAILED", "state":
"stopped" } ],
    "active_incidents": ["INC-0007"]
  }
}
```

A failed request keeps the envelope and carries an error code from the previous section:

```
{
  "ok": false,
  "error": { "code": "SENSOR_NOT_FOUND", "message": "No sensor with ID PRT-
WLV-09" }
}
```

`get_events_since(timestamp)` returns events in order of `seq`:

```
{
  "event_id": "EVT-000451",
  "seq": 88213,
  "timestamp": "2026-10-18T17:30:05Z",
  "type": "FAULT_ACTIVATED",
  "zone_id": "PRT",
  "device_id": "PRT-WLV-01",
  "incident_id": "INC-0007",
  "details": { "fault": "FALSE_READING", "offset": 140.0 }
}
```

Sensor-related event types: FAULT_ACTIVATED, FAULT_CLEARED, STATUS_CHANGED, DEVICE_REPLACED and FAILOVER_REQUESTED. Action events (ACTION_REQUESTED, ACTION_COMPLETED, ACTION_FAILED) use the same shape.

Linked actuators and traffic lights

The 8 actuators below receive commands from the Brain and are fed by the sensors in the registry.

Device ID	Type	Zone	Fed by sensors	Compatible backup
PRT-PMP-01	Pump	Port	PRT-WLV-01, PRT-FLW-01, PRT-PRS-01	PRT-PMP-02
PRT-PMP-02	Pump	Port	PRT-WLV-02	PRT-PMP-01
IND-VLV-01	Valve	Industrial	IND-TMP-01, IND-TMP-02, IND-PRS-01, IND-FLW-01	none
DTN-VLV-01	Valve	Downtown	DTN-WLV-01	none
RES-VLV-01	Valve	Residential	RES-WLV-01, RES-PRS-01, RES-FLW-01	none
DTN-TLT-01	Traffic light	Downtown	DTN-TRF-01	none
DTN-TLT-02	Traffic light	Downtown	DTN-TRF-02	none
RES-TLT-01	Traffic light	Residential	RES-TRF-01	none

Every action on these devices returns one of five execution statuses: PENDING, RUNNING, COMPLETED, FAILED or PARTIAL (for example, a valve that jams at 42%). A command to an unavailable device returns FAILED or REJECTED with `DEVICE_UNAVAILABLE`.

Open decisions

Seven points need an agreement with the Brain and Guardian teams before the schema is frozen.

- [ ] Confirm the sensor list, ID format and units against the challenge specification.

- [ ] Decide who may see `actual_value`, `delta` and `fault`: Guardian and dashboard only, or the Brain too.
- [ ] Choose readings by polling, by push stream, or both, and confirm the intervals.
- [ ] Choose the clock: simulated time with a speed multiplier, or wall-clock time.
- [ ] Confirm that a replacement device gets a new ID with `replaces`, not the old ID.
- [ ] Decide whether Guardian's verdict (for example QUARANTINED) is stored in the sensor record or kept only by Guardian.
- [ ] Agree on the MCP envelope and error codes with the other layers.